

The
HACKER
Ethic

and the Spirit of
the Information Age

PEKKA HIMANEN

Copyright © 2001 Pekka Himanen
Prologue copyright © 2001 Linus Torvalds
Epilogue copyright © 2001 Manuel Castells
Pekka Himanen's text translated by Anselm Hollo and Pekka Himanen
All rights reserved under International and Pan-American
Copyright Conventions. Published in the United States by
Random House Trade Paperbacks, a division of Random House, Inc.,
New York, and simultaneously in Canada by
Random House of Canada Limited, Toronto

RANDOM HOUSE TRADE PAPERBACKS and colophon are
trademarks of Random House, Inc.

This work was originally published in hardcover by
Random House Trade Publishers in January 2001.

Grateful acknowledgment is made to the following for
permission to reprint copyrighted material:

Richard Stallman: "The Free Software Song," by Richard Stallman. Copyright ©
1993 by Richard Stallman. Used by permission. Verbatim redistribution
permitted if this notice is preserved.

Youth Radio, Berkeley, California: E-mails between Finnegan Hamill and
"Adona." Copyright © Youth Radio, Berkeley, California. Used by permission.

Library of Congress Cataloging-in-Publication Data
Himanen, Pekka.

The hacker ethic, and the spirit of the information age / Pekka Himanen.
p. cm.

Includes bibliographical references and index.
ISBN 0-375-75878-X

1. Computer programming—Moral and ethical aspects. 2. Computer hackers.
3. Open source software. I. Title.

QA76.9.M65 H56 2001
174'.90904—dc21 00-053354

Random House website address: www.atrandom.com

Printed in the United States of America on acid-free paper

2 4 6 8 9 7 5 3

First Trade Paperback Edition

The Hacker Work Ethic

Linus Torvalds says in his Prologue that, for the hacker, “the computer itself is entertainment,” meaning that the hacker programs because he finds programming intrinsically interesting, exciting, and joyous.

The spirit behind other hackers’ creations is very similar to this. Torvalds is not alone in describing his work with statements like “Linux hackers do something because they find it to be very interesting.” For example, Vinton Cerf, who is sometimes called “the father of the Internet,” comments on the fascination programming experts: “There was something amazingly enticing about programming.”¹ Steve Wozniak, the person who built the first real personal computer, says forthrightly about his discovery of the wonders of programming: “It was just the most intriguing world.”² This is a general spirit: hackers program because programming challenges are of intrinsic *in-*

terest to them. Problems related to programming arouse genuine curiosity in the hacker and make him eager to learn more.

The hacker is also *enthusiastic* about this interesting thing; it energizes him. From the MIT of the sixties onward, the classic hacker has emerged from sleep in the early afternoon to start programming with enthusiasm and has continued his efforts, deeply immersed in coding, into the wee hours of the morning. A good example of this is the way sixteen-year-old Irish hacker Sarah Flannery describes her work on the so-called Cayley-Purser encryption algorithm: "I had a great feeling of excitement. . . . I worked constantly for whole days on end, and it was exhilarating. There were times when I never wanted to stop."³

Hacker activity is also *joyful*. It often has its roots in playful explorations. Torvalds has described, in messages on the Net, how Linux began to expand from small experiments with the computer he had just acquired. In the same messages, he has explained his motivation for developing Linux by simply stating that "it was/is fun working on it."⁴ Tim Berners-Lee, the man behind the Web, also describes how this creation began with experiments in linking what he called "play programs."⁵ Wozniak relates how many characteristics of the Apple computer "came from a game, and the fun features that were built in were only to do one pet project, which was to program . . . [a game called] Breakout and show it off at the club."⁶

Flannery comments on how her work on the development of encryption technology evolved in the alternation between library study of theorems and the practice of exploratory programming: “With a particularly interesting theorem . . . I’d write a program to generate examples. . . . Whenever I programmed something I’d end up playing around for hours rather than getting back to plodding my way through the paper.”⁷

Sometimes this joyfulness shows in the hacker’s “flesh life” as well. For example, Sandy Lerner is known not only for being one of the hackers behind the Internet routers but also for riding naked on horseback. Richard Stallman, the bearded and longhaired hacker guru, attends computer gatherings in a robe, and he exorcises commercial programs from the machines brought to him by his followers. Eric Raymond, a well-known defender of hacker culture, is also known for his playful lifestyle: a fan of live role-playing games, he roams the streets of his Pennsylvania hometown and the surrounding woods attired as an ancient sage, a Roman senator, or a seventeenth-century cavalier.

Raymond has also given a good summary of the general hacker spirit in his description of the Unix hackers’ philosophy:

To do the Unix philosophy right, you have to be loyal to excellence. You have to believe that software is a craft worth all the intelligence and passion you

can muster. . . . Software design and implementation should be a joyous art, and a kind of high-level play. If this attitude seems preposterous or vaguely embarrassing to you, stop and think; ask yourself what you've forgotten. Why do you design software instead of doing something else to make money or pass the time? You must have thought software was worthy of your passions once. . . .

To do the Unix philosophy right, you need to have (or recover) that attitude. You need to *care*. You need to *play*. You need to be willing to *explore*.⁸

In summing up hacker activity's spirit, Raymond uses the word *passion*, which corresponds to Torvalds's *entertainment*, as he defined it in the Prologue. But Raymond's term is perhaps even more apt because, even though both words have associations that are not meant in this context, *passion* conveys more intuitively than *entertainment* the three levels described above—the dedication to an activity that is intrinsically interesting, inspiring, and joyous.

This passionate relationship to work is not an attitude found only among computer hackers. For example, the academic world can be seen as its much older predecessor. The attitude of passionate intellectual inquiry received similar expression nearly 2,500 years ago when Plato, founder of the first academy, said of philosophy, "Like light flashing forth when a fire is kindled, it is born in the soul and straightway nourishes itself."⁹

The same attitude may also be found in any number of other spheres of life—among artists, artisans, and the “information professionals,” from managers and engineers to media workers and designers, for example. It is not only the hackers’ “jargon file” that emphasizes this general idea of being a hacker. At the first Hacker Conference in San Francisco in 1984, Burrell Smith, the hacker behind Apple’s Macintosh computer, defined the term as follows: “Hackers can do almost anything and be a hacker. You can be a hacker carpenter. It’s not necessarily high tech. I think it has to do with craftsmanship and caring about what you’re doing.”¹⁰ Raymond notes in his guide “How to Become a Hacker” that “there are people who apply the hacker attitude to other things [than software], like electronics and music—actually, you can find it at the highest levels of any science or art.”¹¹

Looked at on this level, computer hackers can be understood as an excellent example of a more general work ethic—which we can give the name *the hacker work ethic*—gaining ground in our network society, in which the role of information professionals is expanding. But although we use a label coined by computer hackers to express this attitude, it is important to note that we could talk about it even without any reference to computer people. We are discussing a general social challenge that calls into question the Protestant work ethic that has long governed our lives and still maintains a powerful hold on us.

Let’s see what type of long historical and strong societal

forces the hacker work ethic, in this sense, faces. The familiar expression “Protestant work ethic” derives, of course, from Max Weber’s famous essay *The Protestant Ethic and the Spirit of Capitalism* (1904–1905).¹² Weber starts out by describing how the notion of work as a duty lies at the core of the capitalist spirit that arose in the sixteenth century: “This peculiar idea, so familiar to us today, but in reality so little a matter of course, of one’s duty in a calling, is what is most characteristic of the social ethic of capitalistic culture, and is in a sense the fundamental basis of it. It is an obligation which the individual is supposed to feel and does feel towards the content of his professional activity, no matter in what it consists, in particular no matter whether it appears on the surface as a utilization of his personal powers, or only of his material possessions (as capital).” Weber goes on to say: “Not only is a developed sense of responsibility absolutely indispensable, but in general also an attitude which, at least during working hours, is freed from continual calculations of how the customary wage may be earned with a maximum of comfort and a minimum of exertion. Labour must, on the contrary, be performed as if it were an absolute end in itself, a calling.”¹³

Then Weber demonstrates how the other main force described in his essay, the work ethic taught by Protestants, which also arose in the sixteenth century, furthered these goals. The Protestant preacher Richard Baxter expressed that work ethic in its pure form: “It is for action that God

maintaineth us and our activities; work is the moral as well as the natural end of power,” and to say “I will pray and meditate [instead of working], is as if your servant should refuse his greatest work and tie himself to some lesser, easier part.”¹⁴ God is not pleased to see people just meditating and praying—he wants them to do their job.

True to the capitalist spirit, Baxter advises employers to reinforce this idea in workers of wanting to do one’s job as well as possible by making it a matter of conscience: “A truly godly servant will do all your service in obedience to God, as if God Himself had bid him do it.”¹⁵ Baxter sums up this attitude by referring to labor as a “calling,”¹⁶ a good expression of the three core attitudes of the Protestant work ethic: work must be seen as an end in itself, at work one must do one’s part as well as possible, and work must be regarded as a duty, which must be done because it must be done.

While the hacker work ethic’s precursor is in the academy, Weber says that the Protestant ethic’s only historical precursor is in the monastery. And certainly, if we expand on Weber’s comparison, we can see many similarities. In the sixth century, for example, Benedict’s monastic rule required all monks to see the work assigned to them as their duty and warned work-shy brethren by noting that “idleness is the enemy of the soul.”¹⁷ Monks were also not supposed to question the jobs they were given. Benedict’s fifth-century predecessor John Cassian made this clear in his monastic rule by describing in admiring tones the obedi-

ence of a monk, named John, to his elder's order to roll a stone so large that no human being could move it:

Again, when some others were anxious to be edified by the example of his [John's] obedience, the elder called him and said: "John, run and roll that stone hither as quickly as possible;" and he forthwith, applying now his neck, and now his whole body, tried with all his might and main to roll an enormous stone which a great crowd of men would not be able to move, so that not only were his clothes saturated with sweat from his limbs, but the stone itself was wetted by his neck; in this too never weighing the impossibility of the command and deed, out of reverence for the old man and the unfeigned simplicity of his service, as he believed implicitly that the old man could not command him to do anything vain or without reason.¹⁸

This Sisyphean straining epitomizes the idea, central to monastic thought, that one should not question the nature of one's work.¹⁹ Benedict's monastic rule even explained that the nature of the work did not matter because the highest purpose of work was not actually to get something done but to *humble* the worker's soul by making him do whatever he is told—a principle that seems to be still active in a great number of offices.²⁰ In medieval times, this prototype for the Protestant work ethic existed only within the monasteries, and it did not influence the prevailing attitude of the church, much less that of society at

large. It was only the Protestant Reformation that spread the monastic thinking to the world beyond the monastery walls.

However, Weber went on to emphasize that even though the spirit of capitalism found its essentially religious justification in the Protestant ethic, the latter soon emancipated itself from religion and began to operate according to its own laws. To use Weber's famous metaphor, it turned into a religiously neutral iron cage.²¹ This is an essential qualification. In our globalizing world, we should think of the term *Protestant ethic* in the same way we think of an expression such as *platonic love*. When we say that someone loves another person platonically, we do not mean that he is a Platonist—that is, an adherent of Plato's philosophy, metaphysics and all. We may attribute a platonic love relationship to a follower of any philosophy, religion, or culture. In the same way, we can speak of someone's "Protestant ethic" regardless of his or her faith or culture. Thus, a Japanese person, an atheist, or a devout Catholic may act—and often does act—in accordance with a Protestant ethic.

One need not look very far to realize how strong a force this Protestant ethic still is. Commonplace remarks like "I want to do my job well," or those made by employers in their little speeches at employee retirement parties about how a person "has always been an industrious/responsible/reliable/loyal worker" are the legacy of the Protestant ethic in that they make no demands on the na-

ture of the work itself. The elevation of work to the status of the most important thing in life—at its extreme, a work addiction that leads to complete neglect of one's loved ones—is another symptom of the Protestant ethic. So is work done with clenched jaws and a responsibility-ridden attitude and the bad conscience many feel when they have to miss work due to ill health.

Seen in a larger historical context, this continued dominance of the Protestant ethic is not so surprising when we remember that even though our network society differs in many significant ways from its predecessor, the industrial society, its “new economy” does not involve a total break with the capitalism Weber describes; it is merely *a new kind of capitalism*. In *The Information Age*, Castells stresses that work, in the sense of labor, is not about to end, despite wild paradisiacal forecasts such as Jeremy Rifkin's *The End of Work*. We easily fall for this illusion that technological advances will, somehow, automatically, make our lives less work-centered—but if we just look at the empirical facts of the rise of the network society so far and project them into the future, we must agree with Castells on the nature of the prevailing pattern: “Work is, and will be for the foreseeable future, the nucleus of people's life.”²² The network society itself does not question the Protestant ethic. Left to its own devices, the work-centered spirit easily continues to dominate within it.

Seen in this overall context, the radical nature of general hackerism consists of its proposing an alternative spirit for the network society—a spirit that finally ques-

tions the dominant Protestant ethic. In this context, we find the only sense in which all hackers are really crackers: they are trying to crack the lock of the iron cage.

The Purpose of Life

The displacement of the Protestant ethic will not happen overnight. It will take time, like all great cultural changes. The Protestant ethic is so deeply embedded in our present consciousness that it is often thought of as if it were just “human nature.” Of course, it is not. Even a brief look at pre-Protestant attitudes toward work provides a healthy reminder of that fact. Both the Protestant and the hacker ethic are historically singular.

Richard Baxter’s view of work was completely alien to the pre-Protestant church. Before the Reformation, clerics tended to devote time to questions such as “Is there life after death?” but none of them worried about whether there was *work* after life. Work did not belong among the church’s highest ideals. God himself worked for six days and finally rested on the seventh. This was the highest goal for human beings as well: in Heaven, just as on Sundays, people would not have to work. Paradise was in, office was out. One might say that Christianity’s original answer to the question “What is the purpose of life?” was: the purpose of life is Sunday.

This statement is not just a witticism. In the fifth century, Augustine compared our life quite literally to Friday, the day when, according to the teachings of the church,

Adam and Eve sinned and Christ suffered on the cross.²³ Augustine wrote that in Heaven we'll find a perennial Sunday, the day on which God rested and Christ ascended to Heaven: "That will truly be the greatest of Sabbaths; a Sabbath that has no evening." Life is just a long wait for the weekend.

Because the Church Fathers saw work as merely a consequence of the fall from grace, they also took very particular conceptual care in their descriptions of Adam's and Eve's activities in Paradise. Whatever Adam and Eve may have done there, it could not be seen as *work*. Augustine emphasizes that in Eden "praiseworthy work was not toilsome"—it was no more than a pleasant *hobby*.²⁴

The pre-Protestant churchmen understood work, "toil," as punishment. In medieval visionary literature that speaks to churchmen's images of Hell, the implements of labor fully reveal their true nature as instruments of torture: sinners are punished with hammers and other tools.²⁵ What's more, according to these visions, there is in Hell an even more cruel torture than the directly inflicted physical one: perennial toil. When the devout brother Brendan saw, in the sixth century, a worker on his visit to the beyond, he immediately made the sign of the cross: he realized that he had arrived where all hope must be abandoned. Here is the narrator of his vision:

When they had passed on further, about a stone's throw, they heard the noise of bellows blowing like

thunder, and the beating of sledge hammers on the anvils and iron. Then St. Brendan armed himself all over his body with the sign of the Cross, saying, “O Lord Jesus Christ, deliver us from this sinister island.” Soon one of the inhabitants appeared to do some work. He was hairy and hideous, blackened with fire and smoke. When he saw the servants of Christ near the island, he withdrew into his forge, crying aloud: “Woe! Woe! Woe!”²⁶

If you do not conduct yourself well in this life, the thinking went, you are condemned to work even in the next. And, even worse, that work, according to the pre-Protestant church, will be absolutely useless, meaningless to an extent you could never have imagined even on your worst working day on earth.

This theme crystallizes in the apotheosis of the pre-Protestant worldview, Dante’s *Divine Comedy* (completed just before his death in 1321), in which sinners who have devoted their lives to money—both spendthrifts and misers—are doomed to push huge boulders around an eternal circle:

More shades were here than anywhere above,
and from both sides, to the sounds of their own screams,
straining their chests, they rolled enormous weights.

And when they met and clashed against each other
they turned to push the other way, one side
screaming, “Why hoard?,” the other side, “Why waste?”

And so they moved back round the gloomy circle,
returning on both sides to opposite poles
to scream their shameful tune another time;

again they came to clash and turn and roll
forever in their semicircle joust.²⁷

Dante borrows this idea from Greek mythology. In Tartarus, where the very worst human beings were dispatched, the most severe punishment was meted out to greedy Sisyphus, who was doomed to endlessly push a big rock up to the top of a hill, from which it always rolled back down.²⁸ Sunday always beckons to Sisyphus and the sinners in Dante's *Inferno*, but it never comes. They are condemned to an eternal Friday.

Considering this background, we can now gain a better understanding of how great a change in our attitude toward work the Protestant Reformation entailed. In allegorical terms, it moved life's center of gravity from Sunday to Friday. The Protestant ethic reoriented ideology so thoroughly that it even turned Heaven and Hell upside down. When work became an end in itself on earth, the clerics found it difficult to imagine Heaven as a place for mere time-wasting leisure, and work could no longer be seen as infernal punishment. Thus, reformed eighteenth-century cleric Johann Kasper Lavater explained that even in Heaven "we cannot be blessed without having occupations. To have an occupation means to have a calling, an office, a special, particular task to do."²⁹ Baptist William

Clarke Ulyat put it in a nutshell when he described Heaven at the beginning of the twentieth century: “practically it is a workshop.”³⁰

The Protestant ethic proved so powerful that its work-centeredness permeated even our imagination. A great example of this is Daniel Defoe’s *Robinson Crusoe* (1719), a novel written by a man trained as a Protestant preacher. Marooned on an abundant island, Crusoe does not take it easy; he works all the time. He is such an orthodox Protestant that he does not even take Sunday off, though he otherwise still observes the seven-day week. After saving an aborigine from his enemies, he aptly names him Friday, trains him in the Protestant ethic, and then praises him in a manner that perfectly describes that ethic’s ideal worker: “Never man had a more faithful, loving, sincere servant, perfectly obliged and engaged; his very affections were ty’d to me, like those of a child to a father.”³¹

In Michel Tournier’s twentieth-century satirical retelling of the novel, *Vendredi* (Friday), Friday’s conversion to the Protestant ethic is still more total. Crusoe decides to put Friday to the test by giving him a task even more Sisyphean than what Cassian’s monastic rule prescribed:

I set him a task which in every prison in the world is held to be the most degrading of harassments—the task of digging a hole and filling it in with the contents of a second; then digging a third, and so on. He la-

bored at this for an entire day, under a leaden sky and in heat like that of a furnace. . . . To say that Friday gave no sign of resenting this idiotic employment, is not enough. I have seldom seen him work with such good will.³²

Sisyphus has truly become a hero.³³

The Passionate Life

When the hacker ethic is placed in this large historical context, it is easy to see that this ethic—understood not just as the computer hackers’ ethic but as a general social challenge—resembles the pre-Protestant ethic to a much greater degree than it does the Protestant one. In this sense, one could say that for hackers the purpose of life is closer to Sunday than to Friday. But, it is important to note, only closer: ultimately, the hacker ethic is not the same as the pre-Protestant work ethic, which envisions a paradise of life without doing anything. Hackers want to realize their passions, and they are ready to accept that the pursuit even of interesting tasks may not always be unmitigated bliss.

For hackers, *passion* describes the general tenor of their activity, though its fulfillment may not be sheer joyful play in all its aspects. Thus, Linus Torvalds has described his work on Linux as a combination of enjoyable hobby and serious work: “Linux has very much been a

hobby (but a serious one: the best type).”³⁴ Passionate and creative, hacking also entails hard work. Raymond says in his guide “How to Become a Hacker”: “Being a hacker is lots of fun, but it’s a kind of fun that takes a lot of effort.”³⁵ Such effort is needed in the creation of anything even just a little bit greater. If need be, hackers are also ready for the less interesting parts necessary for the creation of the whole. However, the meaningfulness of the whole gives even its more boring aspects worth. Raymond writes: “The hard work and dedication will become a kind of intense play rather than drudgery.”³⁶

There’s a difference between being permanently joyless and having found a passion in life for the realization of which one is also willing to take on less joyful but nonetheless necessary parts.